

Higher-order quantum correlations from strictly pairwise interactions: a non-associative algebraic mechanism

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Abstract

The dynamical generation of irreducible higher-order correlations in quantum many-body systems remains poorly understood at the microscopic level. Here we identify the Jordan associator, which encodes the non-associative algebraic residue of incompatible pairwise edge interactions on a quantum network, as an engine for tripartite synergy generation. We decompose this engine into two complementary layers: a bare algebraic capacity J_{tot} determined solely by the Hamiltonian, and a state-activated component J_ρ that measures the projection of the evolving quantum state onto the non-associative sector. In a 4-qubit triangle network, these layers produce a dynamical crossover from an early activation-dominated regime to a later capacity-dominant one. An exact associator-null interpolation demonstrates that suppressing the non-associative channel collapses the tripartite synergy indicator to the numerical floor ($\sim 10^{-13}$) while leaving pairwise correlations intact, establishing non-associativity as a necessary condition for the observed generative route. The recovery reveals a two-layer bottleneck: bare capacity restores first, but synergy generation remains gated by state activation. In a 5-qubit tetrahedral extension, the crossover disappears as self-averaging across multiple triangular faces causes bare capacity to acquire predictive power from the outset — a finite-size effect confirmed independently by initial-state ensemble averaging on the triangle. Validation on the transverse-field Ising model yields an analytic associator $J_{\text{tot}}^2 = h^4 \sum J_{ij}^2$, with synergy peaking near the quantum phase transition and frustrated configurations producing systematically less synergy than non-frustrated ones at fixed capacity. Under amplitude damping, dephasing, and depolarizing noise, the synergy-dominant signal retains a growing share of total correlations relative to pairwise baselines. These results establish a direct link between non-associative interaction algebra and the emergent information-theoretic organization of quantum networks, and point toward a Jordan-algebraic extension of spectral network theory.

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I. INTRODUCTION

Higher-order interactions are increasingly recognized as an essential organizing principle of complex systems, and have given rise to a rich theory built on hypergraphs and simplicial complexes [1]. Within this framework, information-theoretic diagnostics such as the O-information [2] and its gradient-based extensions [3] characterize synergy- and redundancy-dominated structure, while dynamical models with explicit multi-body couplings have revealed phenomena—including explosive synchronization [4] and discontinuous contagion transitions [5]—that pairwise descriptions cannot capture. A complementary line of work has identified mechanisms by which higher-order signatures emerge from strictly pairwise interactions: nonlinear phase reduction with frequency resonance produces effective three-body terms in coupled oscillators [6], antibalanced correlational structures generate synergy-dominance in Gaussian systems [7], and explicit group-level mechanisms produce distinctive synergistic signatures in classical higher-order networks [8]. Together, these results establish that higher-order organization in classical complex systems can be traced either to explicit higher-order coupling terms or to statistical, structural, and dynamical properties of pairwise interactions.

A natural further question is whether the *algebraic* structure of the pairwise interactions themselves can act as a generator of higher-order correlations. This question concerns a level of structure distinct from the multivariate functional content of node-level dynamics: graph-based formulations admit arbitrarily complex node functions and, through appropriate reparametrization, can express the dynamics of any hypergraph model [9]. Such reparametrizations, however, operate within the commutative algebra of classical observables and do not generate non-trivial associator structure among pairwise interaction terms. The algebraic question therefore probes a level—the non-commutative algebra of interaction operators—that is absent from classical networks with scalar observables. Quantum mechanics provides this level natively: while the Jordan product $X \circ Y = (XY + YX)/2$ of Hermitian operators is commutative, it is generally non-associative, and its associator captures a genuine three-operator incompatibility whose existence depends on the operator-algebraic structure rather than on the choice of node-level dynamical functions. Quantum tripartite mutual information and related observables have been used to probe scrambling [10–12], post-quench dynamics [13], and higher-order interdependencies in entangled states [14]. Yet

these studies predominantly characterize the *presence* and *magnitude* of higher-order structure at the level of quantum states, leaving unidentified [14] the specific Hamiltonian-level property that determines whether and how efficiently irreducible tripartite correlations are generated from initially unentangled states.

Here we show that the answer to the algebraic version of this question lies in the non-associative structure of the Hamiltonian’s Jordan product. We identify the Jordan associator, which encodes the irreducible three-edge interference residue of pairwise interactions, as a necessary engine for the dynamical generation of synergy-dominant tripartite correlations. The generation is governed by two complementary layers: a state-activated component J_ρ that quantifies how strongly the evolving state projects onto that sector, and a bare algebraic capacity J_{tot} that quantifies the available non-associative structure. In a 4-qubit triangle network, these layers produce a dynamical crossover from an early activation-dominated regime to a later capacity-dominant one. By constructing an exact associator-null manifold, we establish that suppressing the non-associative channel collapses the tripartite signal to the numerical floor while leaving pairwise correlations intact, identifying non-associativity as a necessary condition for the observed generative route. In a 5-qubit tetrahedral extension, bare capacity becomes informative at earlier times, revealing the crossover as a finite-size effect governed by self-averaging across triangular faces.

II. FRAMEWORK

A. System setup

We consider simplex-based quantum networks built from fixed-norm XYZ edge Hamiltonians,

$$H_{ij} = \sum_{\mu=x,y,z} J_{ij}^{(\mu)} \sigma_i^\mu \sigma_j^\mu. \quad (1)$$

The minimal system is a triangle A – B – C coupled to an auxiliary node D through a single weak edge H_{CD} , with total Hamiltonian

$$H^{(4)} = H_{AB} + H_{BC} + H_{AC} + H_{CD}. \quad (2)$$

The auxiliary node D plays a structurally necessary role: for any pure tripartite state ρ_{ABC} , the Schmidt duality $S_A = S_{BC}$, $S_B = S_{AC}$, $S_C = S_{AB}$ forces the von Neumann counterpart

of I_3 to vanish identically [14]. Nontrivial tripartite information generation therefore requires ρ_{ABC} to be mixed, which the present construction realizes by tracing out D from a closed four-qubit evolution. The four-qubit triangle is in this sense the minimal closed system in which dynamical generation of synergy-dominant tripartite correlations from product initial states can be observed.

The tetrahedral extension replaces the internal triangle with the complete graph on A, B, C, D (six edges), while coupling an external node E through a single edge H_{DE} . Starting from a product initial state ρ_0 , we evolve the full system unitarily and evaluate observables on the reduced internal state,

$$\rho_{ABC}(t) = \text{Tr}_D \left[e^{-iH^{(4)}t} \rho_0 e^{iH^{(4)}t} \right], \quad (3)$$

and analogously $\rho_{ABCD}(t) = \text{Tr}_E[\dots]$ in the tetrahedral case. In both geometries, the internal subsystem evolves under fixed-norm interactions, while the external node provides the source of mixedness required by the minimality argument above and is traced out after evolution. Full details of the Hamiltonian parametrization, initial-state construction, and scanning protocol are given in Appendix ??.

While our primary numerical experiments employ the XYZ Hamiltonian family, which provides a continuously tunable non-associative structure, the algebraic framework applies to any Hamiltonian with pairwise interactions. As a concrete example, we verify in Appendix ?? that the transverse-field Ising model $\sigma_i^z \sigma_j^z + h(\sigma_i^x + \sigma_j^x)/2$ yields an analytic associator $J_{\text{tot}}^2 = h^4 \sum_{(ij)} J_{ij}^2$, which vanishes in the pure Ising limit $h \rightarrow 0$ and is activated by the transverse field. This confirms that the non-associative generation channel is not an artifact of the XYZ parametrization but arises generically when incompatible interaction axes coexist.

B. Algebraic motivation

For a tripartite reduced state ρ_{ABC} , we define the tripartite synergy indicator

$$I_3^-(A:B:C) \equiv \max(0, -I_3(A:B:C)), \quad (4)$$

where

$$I_3(A:B:C) = S_A + S_B + S_C - S_{AB} - S_{AC} - S_{BC} + S_{ABC} \quad (5)$$

is the tripartite mutual information and $S_X = -\text{Tr}[\rho_X \log_2 \rho_X]$ denotes the von Neumann entropy. For three variables, I_3 coincides with the McGill interaction information and, when

entropies are evaluated as von Neumann entropies, with the Q-information restricted to three variables [2, 14]; $I_3^- > 0$ corresponds to its synergy-dominant branch, where irreducible synergistic dependencies exceed redundant ones, while $I_3^- = 0$ indicates a redundancy-dominated or balanced regime. The associated decomposition into unique, redundant, and synergistic contributions in the quantum setting remains an open theoretical challenge, addressed further in the Discussion.

To probe the Hamiltonian-level origin of I_3^- , we introduce the mediator Jordan associator for each edge pair sharing a common face:

$$A_{ij} = \frac{1}{4} [H_{ij}, [H_{ik}, H_{jk}]], \quad (6)$$

which is the nested-commutator representation of the Jordan associator $[H_{ik}, H_{ij}, H_{jk}]_\circ = (H_{ik} \circ H_{ij}) \circ H_{jk} - H_{ik} \circ (H_{ij} \circ H_{jk})$ under the Jordan product $A \circ B = \frac{1}{2}(AB + BA)$. This identity is verified symbolically in Appendix ???. As a nested commutator, A_{ij} admits two elementary algebraic vanishing modes: it vanishes when the inner commutator $[H_{ik}, H_{jk}]$ is zero because the edge operators on that face commute pairwise (*alignment-driven vanishing*), and it vanishes when one of the participating edge operators has shrinking norm (*magnitude-driven vanishing*). Both modes correspond to the operator algebra failing to span the three-edge configuration non-trivially, and both are tested against I_3^- in Sec. ???.

From A_{ij} we construct two scalar diagnostics. The *bare algebraic capacity*

$$J_{\text{tot}} = \sqrt{\sum_{(ij)} \|A_{ij}\|_F^2} \quad (7)$$

depends only on the Hamiltonian and quantifies the total non-associative structure available on a given face. The *state-activated diagnostic*

$$J_\rho = \sqrt{\sum_{(ij)} |\text{Tr}[\rho A_{ij}]|^2} \quad (8)$$

additionally depends on the quantum state, measuring how strongly the evolving reduced state projects onto the non-associative sector. A Cauchy–Schwarz argument yields the bound $J_\rho/J_{\text{tot}} \leq \sqrt{\text{Tr}[\rho^2]}$, linking the activation efficiency to the purity of the reduced state (Appendix A).

The associator-based diagnostics are not chosen ad hoc. In the short-time perturbative expansion of the reduced dynamics, the $\mathcal{O}(t^2)$ contribution to the purity $\text{Tr}[\rho^2]$ contains no three-edge cross-terms; genuine three-edge interference enters at $\mathcal{O}(t^3)$ or higher (Appendix A). The Jordan associator captures precisely this irreducible three-edge kernel,

Time-resolved associator diagnostics in the triangle network (4-qubit)
(Fixed random product initial state)

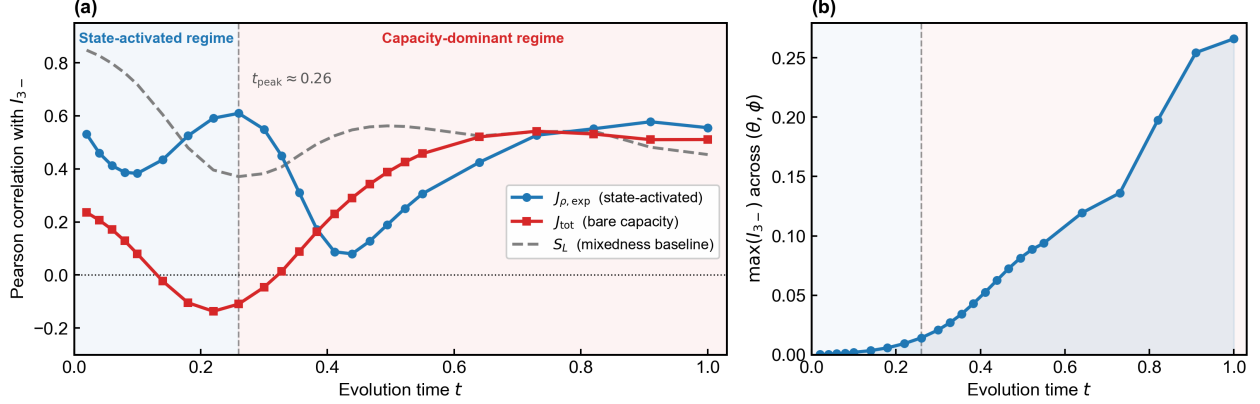


FIG. 1. Time-resolved associator diagnostics in the triangle network (4-qubit). (a) Pearson correlation of each diagnostic with I_3^- as a function of evolution time. The state-activated diagnostic J_ρ (blue) peaks at $t_{\text{peak}} \approx 0.26$ and subsequently declines, while the bare capacity J_{tot} (red) rises monotonically. The dashed grey line shows the linear entropy S_L as a mixedness baseline. (b) Maximum I_3^- across the (θ, ϕ) scan at each time point.

providing a natural Hamiltonian-level candidate for the generator of the tripartite synergy indicator.

III. RESULTS

A. Dynamical crossover in the triangle network

We begin with the 4-qubit triangle network and ask how the associator diagnostics relate to the emergence of synergy-dominant structure over time. For each evolution time t , we scan the full (θ, ϕ) Hamiltonian family and compute the Pearson correlation of I_3^- with J_ρ , J_{tot} , and the linear entropy $S_L = 1 - \text{Tr}[\rho_{ABC}^2]$ as a mixedness baseline (Fig. 1).

A clear dynamical crossover emerges (Fig. 1a). At early times, J_ρ is the strongest correlate of I_3^- : the generation of synergy-dominant structure is initially governed by whether the evolving state activates the non-associative sector. At later times, J_ρ declines while J_{tot} rises monotonically, indicating that the bare algebraic capacity—a state-independent property of the Hamiltonian—becomes the dominant predictor as the system thermalizes.

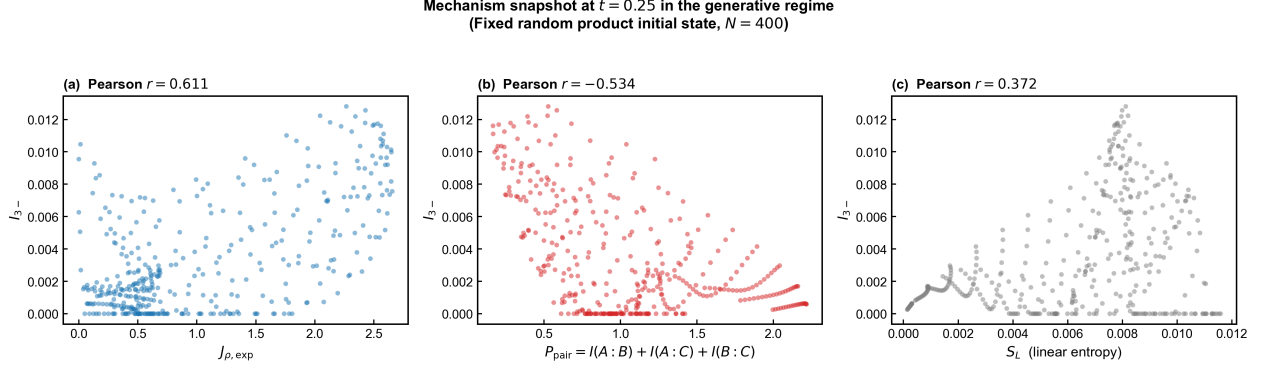


FIG. 2. **Cross-sectional evidence at $t = 0.25$.** (a) J_ρ vs. I_3^- : positive correlation ($r = 0.61$). (b) P_{pair} vs. I_3^- : negative correlation ($r = -0.53$), with $N_{\text{sum}} \equiv 0$ across all 400 configurations. (c) S_L vs. I_3^- : weaker correlation ($r = 0.37$) with a hook shape indicating that mixedness is necessary but not sufficient.

The linear entropy S_L is strongly correlated at very early times, when all correlations grow in lockstep with the departure from purity, but its predictive power drops below that of J_ρ in the generative window ($t \approx 0.15$ – 0.30), confirming that mixedness growth is a kinematic threshold rather than a mechanistic driver.

To sharpen this picture, Fig. 2 presents cross-sectional scatter plots at $t = 0.25$, within the state-activated regime. Across the scanned Hamiltonian family, J_ρ organizes the high- I_3^- region most clearly ($r = 0.61$), while S_L shows a characteristic hook shape ($r = 0.37$): low mixedness implies low synergy (necessary condition), but high mixedness does not guarantee high synergy (not sufficient). Strikingly, the pairwise negativity vanishes identically across all 400 parameter points ($N_{\text{sum}} \equiv 0$), yet I_3^- reaches values up to 0.013. This demonstrates that synergy-dominant tripartite structure can emerge in the complete absence of detectable bipartite entanglement.

B. Non-associativity as a necessary condition

The Jordan associator $\mathcal{A}_{ij} = \frac{1}{4}[H_{ij}, [H_{ik}, H_{jk}]]$ vanishes identically whenever its inner commutator $[H_{ik}, H_{jk}]$ does. Two elementary algebraic conditions force this inner commutator to zero: alignment of all edge operators within a common commuting subalgebra, and shrinking of an edge operator’s norm to zero. We use the alignment-driven condition to construct a continuously controllable null family; the magnitude-driven condition is realized

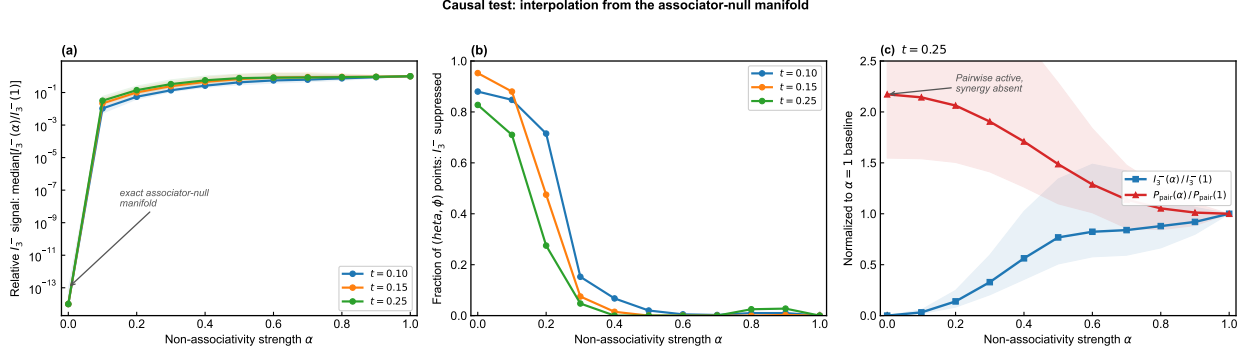


FIG. 3. **Causal test: interpolation from the associator-null manifold.** (a) Median $I_3^-(\alpha)/I_3^-(1)$ on a logarithmic scale: the signal collapses to the numerical floor ($\sim 10^{-13}$) at $\alpha = 0$ and recovers monotonically. (b) Fraction of (θ, ϕ) points at which pairwise correlations are retained ($P_{\text{pair}} > 50\%$) but I_3^- is suppressed ($< 10\%$ of its $\alpha = 1$ value). (c) Contrasting recovery of I_3^- (blue) and P_{pair} (red), both normalized to $\alpha = 1$, at $t = 0.25$.

within the transverse-field Ising specialization (Appendix E).

A natural exact realization of the alignment condition is the pure Ising family ($\sigma_i^z \sigma_j^z$): all edge operators share the computational diagonal basis, pairwise commute, and the associator vanishes identically. Numerical verification confirms that $I_3^- = 0$ to machine precision under pure Ising evolution despite active pairwise correlations (Appendix B 5). To continuously interpolate between this null and the full XYZ family, we construct an exact associator-null control family. Each fixed-norm XYZ edge Hamiltonian is continuously deformed toward a common-axis configuration in which all three internal edge operators commute pairwise; at this null point the Jordan associators vanish identically while edge norms, and hence pairwise interaction strengths, are preserved. An interpolation parameter $\alpha \in [0, 1]$ connects the null manifold ($\alpha = 0$) to the original XYZ family ($\alpha = 1$); full construction details are given in Appendix B.

The results are unambiguous (Fig. 3). At $\alpha = 0$, I_3^- collapses to the numerical floor (median $\sim 10^{-13}$), while pairwise backgrounds remain robustly active: the median P_{pair} and N_{pair} even exceed their $\alpha = 1$ values (Fig. 3c). This confirms that the null manifold suppresses the higher-order signal not by weakening the overall interaction scale, but by specifically eliminating the non-associative three-edge channel. As α increases, I_3^- recovers monotonically, and the fraction of parameter points exhibiting strong suppression decays

from near unity at $\alpha = 0$ to zero at $\alpha = 1$ (Fig. 3b). The recovery itself reveals a two-layer bottleneck: J_{tot} restores rapidly at small α , but I_3^- recovers much more slowly, closely tracked by J_ρ , which confirms that non-associative capacity is necessary but not sufficient, and that state activation is the rate-limiting step (Appendix C).

The complementary algebraic route to the same vanishing is realized in the transverse-field Ising specialization through the weight-product factor $C_\Delta = 27 w_{AB} w_{BC} w_{AC}$, which controls the dependence of I_3^- on the within-face edge-weight distribution at fixed face capacity (Appendix E). Synergy-dominant generation collapses along this magnitude-driven route as well. Together, the alignment-driven and magnitude-driven vanishing modes exhaust the elementary algebraic conditions under which the Jordan associator trivializes, and both confirm that non-associativity is a necessary and continuously tunable ingredient for the dynamical generation of synergy-dominant tripartite structure.

C. Tetrahedral extension

Does the two-layer picture persist in a richer topology? We extend the construction to a 5-qubit tetrahedral core A, B, C, D coupled to an external node E . The reduced state ρ_{ABCD} hosts four triangular faces, on each of which we evaluate I_3^- and the associator diagnostics, reporting face-averaged quantities (Fig. 4).

The qualitative architecture of the two-layer framework remains intact: J_ρ retains a well-defined peak marking the optimal window of state-resolved diagnostics, while J_{tot} provides a time-robust structural predictor. However, a quantitative shift is evident: J_{tot} becomes informative at much earlier times than in the triangle, while J_ρ no longer dominates at any time point.

This shift is consistent with a self-averaging mechanism. The tetrahedron averages over four triangular faces whose individual non-associative capacities vary with the Hamiltonian parametrization. This averaging suppresses the fluctuations that mask the bare-capacity signal in the minimal triangle, allowing structural information to become visible earlier—in direct analogy to how mean-field descriptions sharpen as the number of interacting subsystems increases. A complementary signature of the same fluctuation-suppression principle appears in an initial-state ensemble on the 4-qubit triangle: averaging over 50 random product states likewise causes J_{tot} to become informative earlier than in any single-state

Time-resolved associator diagnostics in the tetrahedral network (5-qubit)
(Fixed random product initial state, $g_E = 0.2$)

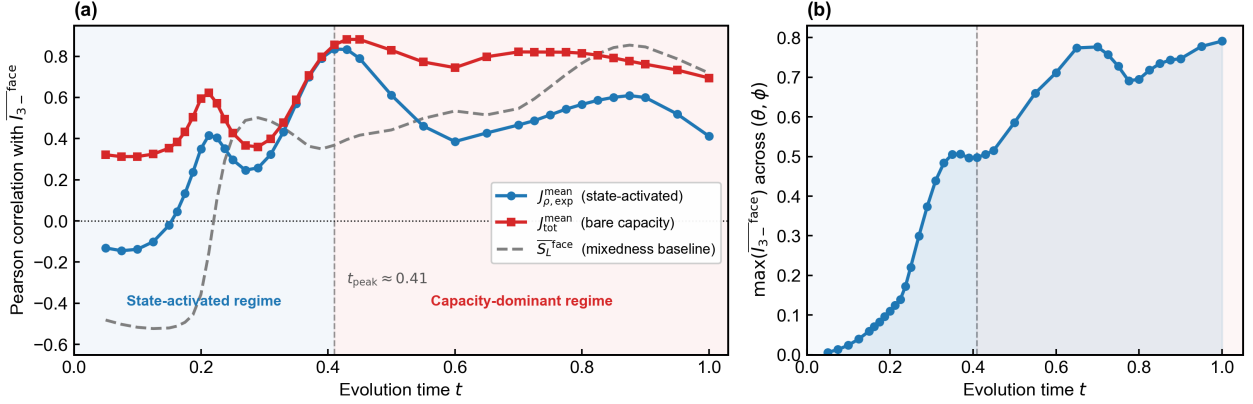


FIG. 4. Time-resolved associator diagnostics in the tetrahedral network (5-qubit).

Same format as Fig. 1. The bare capacity $J_{\text{tot}}^{\text{mean}}$ (red) is informative from early times; J_{ρ}^{mean} (blue) retains a peak at $t_{\text{peak}} \approx 0.41$ but no longer dominates.

scan (Appendix ??). That the same mechanism operates across two independent averaging channels—face averaging in the tetrahedron and state-ensemble averaging in the triangle—strengthens the interpretation of self-averaging as a general statistical feature of the two-layer framework.

IV. DISCUSSION

Our results establish the non-associative algebraic residue of pairwise quantum interaction, quantified by the Jordan associator, as a necessary and continuously tunable engine for the dynamical generation of synergy-dominant tripartite correlations. The state-activated diagnostic J_{ρ} and the bare algebraic capacity J_{tot} provide complementary predictors whose relative dominance shifts with network scale: in the minimal triangle, J_{ρ} leads at early times and J_{tot} at late times; in the tetrahedron, self-averaging across multiple faces causes J_{tot} to acquire predictive power from the outset. The associator-null manifold provides a continuous causal control: suppressing the non-associative channel collapses the tripartite signal to the numerical floor while leaving pairwise correlations intact. Under amplitude damping, dephasing, and depolarizing noise, the synergy-dominant signal retains a growing share of total correlations relative to pairwise baselines in both geometries (Appendix F).

This mechanism complements previously identified classical channels for higher-order emergence—phase-resonance reduction [6], antibalanced correlational structure [7, 15], and explicit group-level couplings [8]—by operating at the level of the operator algebra formed by pairwise interactions rather than at the level of correlation structure, dynamical resonance, or coupling architecture. Because the Jordan associator is a nested-commutator residue, it lies outside the scope of any reparametrization within commutative observable algebras [9], identifying non-associativity as a mechanism class that classical networks with scalar observables structurally cannot host. Whether analogous constructions arise in classical systems whose local degrees of freedom carry non-commutative algebraic structure is an open question that this work motivates.

Validation on the transverse-field Ising model yields an analytic capacity $J_{\text{tot}}^2 = h^4 \sum_{(ij)} J_{ij}^2$. The generated synergy peaks at an intermediate field, consistent with a finite-size competition window between Ising interactions and transverse-field dynamics (Appendix E). At fixed capacity, frustrated configurations produce less synergy than non-frustrated ones systematically, confirming that frustration suppresses state activation J_ρ rather than algebraic capacity—a dynamical effect distinct from the equilibrium synergy enhancement reported in classical Ising models [15].

The disappearance of the early-time crossover in the tetrahedron, together with analogous fluctuation suppression across initial-state ensembles in the triangle (Appendix D), is consistent with a self-averaging effect across independent triangular channels. In the TF-Ising case, the bare capacity is graph-additive across triangular faces, so that face-mean fluctuations scale as $1/\sqrt{N}$ on triangle-decorated lattices (Appendix E5). Whether J_{tot} alone suffices as a structural predictor in the thermodynamic limit is a natural target for larger- N extensions.

Several limitations should be noted. First, I_3^- quantifies the net synergy–redundancy balance, not the absolute magnitude of synergistic information; a full decomposition would require partial information decomposition [2, 3] adapted to quantum density matrices, which remains an open challenge. Complementary diagnostics built from inclusion–exclusion of entanglement entropies [16] offer an alternative window whose relation to the associator mechanism merits exploration. Second, the single auxiliary node used to induce reduced-state mixedness is a minimal environment model; extension to richer environment structures remains to be carried out. Third, capacity and activation are necessary but not jointly suf-

ficient: even at fixed J_{tot} , imbalanced edge-weight distributions within a triangular face suppress synergy generation (Appendix ??), motivating a spectral extension of the framework.

Looking forward, the most direct extension is a Jordan Laplacian—a spectral operator built from face-level associator weights—that would connect the present microscopic diagnostics to the spectral theory of higher-order networks [1, 17]; the graph-additive structure of J_{tot}^2 provides a concrete starting point for such a construction. Our results suggest that the boundary between pairwise and higher-order physics is permeable in quantum systems, mediated by the non-associative character of the operator algebra.

Appendix A: Algebraic framework and theoretical bounds

This appendix collects the algebraic identities used in the main text and clarifies the relation between the two associator diagnostics, J_{tot} and J_ρ . The purpose is not to derive a closed formula for the tripartite synergy indicator I_3^- , but to identify the Hamiltonian-level algebraic kernel that is probed by the numerical experiments.

1. Jordan associator identity

The interaction terms considered in this work are Hermitian edge Hamiltonians. For two Hermitian operators X and Y , the ordinary operator product XY is not generally Hermitian. By contrast, the Jordan product

$$X \circ Y = \frac{1}{2}(XY + YX) \tag{A1}$$

maps Hermitian operators to Hermitian operators. It therefore provides a natural symmetric product at the level of quantum observables.

We use the corresponding Jordan associator to probe whether three edge interactions combine associatively at the observable level:

$$[X, Y, Z]_\circ = (X \circ Y) \circ Z - X \circ (Y \circ Z). \tag{A2}$$

This should not be viewed as the introduction of an independent non-associative algebraic model. Rather, it is the non-associative residue induced by applying the Jordan product to edge Hamiltonians inside the ordinary operator algebra.

Using $X \circ Y = (XY + YX)/2$, the two terms are

$$(X \circ Y) \circ Z = \frac{1}{4}(XYZ + YXZ + ZXY + ZYX), \quad (\text{A3})$$

and

$$X \circ (Y \circ Z) = \frac{1}{4}(XYZ + XZY + YZX + ZYX). \quad (\text{A4})$$

Taking the difference gives

$$[X, Y, Z]_{\circ} = \frac{1}{4}(YXZ + ZXY - XZY - YZX), \quad (\text{A5})$$

or equivalently

$$[X, Y, Z]_{\circ} = \frac{1}{4}[Y, [X, Z]]. \quad (\text{A6})$$

Thus the Jordan associator is a nested-commutator residue of the underlying operator algebra. It vanishes when the relevant operator incompatibilities vanish, and becomes nonzero only through a genuine three-operator incompatibility.

In the numerical implementation, Eq. (A6) was also checked using noncommutative symbolic variables in `SymPy`. This symbolic check is only an algebraic consistency verification; the identity itself follows directly from the expansion above.

2. Face-resolved associator diagnostics

For a triangular face $f = (A, B, C)$, let the three edge Hamiltonians be H_{AB} , H_{BC} , and H_{AC} . We define three mediator-resolved Jordan associator channels by placing each edge once in the middle argument:

$$\mathcal{A}_{AB} = [H_{BC}, H_{AB}, H_{AC}]_{\circ} = \frac{1}{4}[H_{AB}, [H_{BC}, H_{AC}]], \quad (\text{A7})$$

$$\mathcal{A}_{BC} = [H_{AB}, H_{BC}, H_{AC}]_{\circ} = \frac{1}{4}[H_{BC}, [H_{AB}, H_{AC}]], \quad (\text{A8})$$

$$\mathcal{A}_{AC} = [H_{AB}, H_{AC}, H_{BC}]_{\circ} = \frac{1}{4}[H_{AC}, [H_{AB}, H_{BC}]]. \quad (\text{A9})$$

The labels indicate the mediator edge. The precise ordering convention is not essential; the three channels sample the same triangular incompatibility kernel with different mediator choices.

The state-independent bare associator capacity is

$$J_{\text{tot}} = (\|\mathcal{A}_{AB}\|_F^2 + \|\mathcal{A}_{BC}\|_F^2 + \|\mathcal{A}_{AC}\|_F^2)^{1/2}, \quad (\text{A10})$$

where $\|\cdot\|_F$ is the Frobenius norm. This quantity depends only on the face Hamiltonians.

For a reduced face state $\rho_f(t)$, the state-resolved associator readout is

$$J_\rho(t) = (|\text{Tr}[\rho_f(t)\mathcal{A}_{AB}]|^2 + |\text{Tr}[\rho_f(t)\mathcal{A}_{BC}]|^2 + |\text{Tr}[\rho_f(t)\mathcal{A}_{AC}]|^2)^{1/2}. \quad (\text{A11})$$

Thus J_{tot} measures the operator-level size of the associator sector, while J_ρ measures how strongly the evolving state reads out that sector.

3. Cauchy–Schwarz activation bound

The relation between J_ρ and J_{tot} follows from the Hilbert–Schmidt Cauchy–Schwarz inequality. For each mediator channel $\mu \in \{AB, BC, AC\}$,

$$|\text{Tr}[\rho_f\mathcal{A}_\mu]| \leq \|\rho_f\|_F \|\mathcal{A}_\mu\|_F. \quad (\text{A12})$$

Squaring and summing over the three channels gives

$$J_\rho^2 \leq \|\rho_f\|_F^2 J_{\text{tot}}^2 = \text{Tr}[\rho_f^2] J_{\text{tot}}^2. \quad (\text{A13})$$

Hence

$$J_\rho \leq \sqrt{\text{Tr}[\rho_f^2]} J_{\text{tot}}. \quad (\text{A14})$$

Equivalently, for $J_{\text{tot}} > 0$, the normalized activation ratio

$$\eta = \frac{J_\rho}{J_{\text{tot}}} \quad (\text{A15})$$

satisfies

$$\eta \leq \sqrt{\text{Tr}[\rho_f^2]} \leq 1. \quad (\text{A16})$$

This bound implies $J_{\text{tot}} = 0 \Rightarrow J_\rho = 0$ for every state ρ_f . The converse need not hold: a Hamiltonian may contain a nonzero associator sector while the evolving state has little projection onto it. This is the algebraic basis for treating J_{tot} and J_ρ as two levels of the same diagnostic: available capacity and state-conditioned activation.

4. Short-time structure and interpretation

The Cauchy–Schwarz bound contains the scalar factor $\sqrt{\text{Tr}[\rho_f^2]}$, which measures the Frobenius norm of the reduced state. This factor should not be confused with the associator channel itself. To see this, consider the short-time expansion

$$\rho(t) = \rho_0 - it[H, \rho_0] - \frac{t^2}{2}[H, [H, \rho_0]] + \mathcal{O}(t^3), \quad (\text{A17})$$

with

$$H = H_{AB} + H_{BC} + H_{AC} + \cdots. \quad (\text{A18})$$

The $\mathcal{O}(t^2)$ contribution to any reduced-state purity $\text{Tr}[\rho_f(t)^2]$ is at most quadratic in the Hamiltonian terms. It may contain single-edge and two-edge cross terms, but it cannot contain a monomial involving all three internal edges H_{AB} , H_{BC} , and H_{AC} simultaneously. This absence of three-edge terms at order t^2 was verified symbolically in the non-commutative expansion.

Therefore the purity prefactor in Eq. (A16) does not by itself encode the three-edge associator kernel. The associator enters through the expectation values $\text{Tr}[\rho_f(t)\mathcal{A}_\mu]$, where Eq. (A6) identifies $\mathcal{A}_\mu$ as a nested-commutator residue of the edge Hamiltonians. Thus J_{tot} measures the available operator-level three-edge incompatibility on a face, while J_ρ measures the state-conditioned readout of that incompatibility.

This motivates J_ρ as the relevant early-time coordinate for the negative tripartite-information signal: the Hamiltonian may contain a nonzero associator sector, but the reduced state must project onto it for the signal to appear. The identities above should not be read as a closed-form derivation of I_3^- from the associator. They identify the Hamiltonian-level algebraic kernel whose state-conditioned activation is tested in the parameter scans and in the associator-null ablation experiment.

Appendix B: Hamiltonian construction and numerical details

This section specifies the numerical conventions used throughout the scans: the fixed-norm XYZ Hamiltonian family, edge-specific shape vectors, auxiliary coupling convention, initial states, scan grids, time evolution, and associator-null interpolation.

1. Fixed-norm XYZ Hamiltonian family

All XYZ scans use two-qubit edge Hamiltonians

$$H_{ij} = J_x^{(ij)}\sigma_x^i\sigma_x^j + J_y^{(ij)}\sigma_y^i\sigma_y^j + J_z^{(ij)}\sigma_z^i\sigma_z^j, \quad (\text{B1})$$

with coupling vector strength fixed by

$$\|\mathbf{J}_{ij}\| = \sqrt{(J_x^{(ij)})^2 + (J_y^{(ij)})^2 + (J_z^{(ij)})^2} = R_{ij}. \quad (\text{B2})$$

Thus the scan changes the relative XX, YY , and ZZ weights, but not the coefficient-vector norm of each edge.

The two-parameter family is generated from the angular seed

$$\hat{\mathbf{u}}(\theta, \phi) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta), \quad \theta, \phi \in [0, \pi/2], \quad (\text{B3})$$

which is deformed by an edge-specific shape vector $\mathbf{s}_{ij}$:

$$\mathbf{v}_{ij}(\theta, \phi) = \hat{\mathbf{u}}(\theta, \phi) \odot \mathbf{s}_{ij}. \quad (\text{B4})$$

Here $\odot$ denotes componentwise multiplication. The deformed vector is then normalized back to the target edge norm,

$$\mathbf{J}_{ij}(\theta, \phi) = R_{ij} \frac{\mathbf{v}_{ij}(\theta, \phi)}{\|\mathbf{v}_{ij}(\theta, \phi)\|} = R_{ij} \frac{\hat{\mathbf{u}}(\theta, \phi) \odot \mathbf{s}_{ij}}{\|\hat{\mathbf{u}}(\theta, \phi) \odot \mathbf{s}_{ij}\|}. \quad (\text{B5})$$

All edges therefore share the same global (θ, ϕ) scan, while the shape vectors make the edge Hamiltonians non-identical. In the numerical grids below we use a small offset $\theta_{\min} = 0.1$ to avoid the degenerate pole at $\theta = 0$.

For the 4-qubit triangle-plus-auxiliary system,

$$E_{4Q} = \{AB, BC, AC, CD\}, \quad (\text{B6})$$

with

$$\mathbf{s}_{AB} = (1.00, 0.70, 0.30), \quad \mathbf{s}_{BC} = (0.80, 1.00, 0.50), \quad (\text{B7})$$

$$\mathbf{s}_{AC} = (0.60, 0.40, 1.00), \quad \mathbf{s}_{CD} = (1.00, 0.80, 0.60). \quad (\text{B8})$$

For the 5-qubit tetrahedral extension,

$$E_{5Q} = \{AB, AC, AD, BC, BD, CD, DE\}, \quad (\text{B9})$$

with

$$\mathbf{s}_{AB} = (1.00, 0.70, 0.30), \quad \mathbf{s}_{AC} = (0.60, 0.40, 1.00), \quad (\text{B10})$$

$$\mathbf{s}_{AD} = (0.90, 0.55, 0.75), \quad \mathbf{s}_{BC} = (0.80, 1.00, 0.50), \quad (\text{B11})$$

$$\mathbf{s}_{BD} = (0.70, 0.95, 0.65), \quad \mathbf{s}_{CD} = (1.10, 0.85, 0.60), \quad (\text{B12})$$

$$\mathbf{s}_{DE} = (1.00, 0.80, 0.60). \quad (\text{B13})$$

The edge norms R_{ij} , including the weak auxiliary-edge norms, are specified below.

2. Auxiliary coupling

The internal interaction strength is fixed to $R = \sqrt{3}$. In the 4-qubit triangle benchmark, the internal subsystem is ABC , and the auxiliary node D is coupled through a single weak edge CD :

$$H^{(4)} = H_{AB} + H_{BC} + H_{AC} + H_{CD}. \quad (\text{B14})$$

The internal edges have norm

$$R_{AB} = R_{BC} = R_{AC} = R, \quad R_{CD} = g_D R, \quad g_D = 0.20. \quad (\text{B15})$$

The auxiliary node is included in the full unitary evolution, but triangle observables are evaluated on

$$\rho_{ABC}(t) = \text{Tr}_D[\rho(t)]. \quad (\text{B16})$$

Thus D provides a weak environment-like coupling without being part of the target tripartite subsystem.

In the 5-qubit tetrahedral extension, the internal subsystem is the tetrahedron $ABCD$, and the external node E is coupled only through the edge DE :

$$H^{(5)} = H_{AB} + H_{AC} + H_{AD} + H_{BC} + H_{BD} + H_{CD} + H_{DE}. \quad (\text{B17})$$

The six internal tetrahedral edges have norm R , while

$$R_{DE} = g_E R, \quad g_E = 0.20 \quad (\text{B18})$$

for the main time-resolved tetrahedral scan. Stronger values of g_E are used only in the supplementary environment-coupling scan.

The external node E is included in the full evolution and traced out before evaluating internal tetrahedral observables:

$$\rho_{ABCD}(t) = \text{Tr}_E[\rho(t)]. \quad (\text{B19})$$

Face-level tripartite quantities are then computed on

$$ABC, \quad ABD, \quad ACD, \quad BCD. \quad (\text{B20})$$

3. Initial states

In the 4-qubit triangle system, we use three types of initial states. The axis-biased product-state control is

$$|\psi_0^+\rangle = |+\rangle_A \otimes |+\rangle_B \otimes |+\rangle_C \otimes |0\rangle_D, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \quad (\text{B21})$$

This state is unentangled but has an explicit single-axis bias.

The main 4-qubit benchmark uses a fixed random product state,

$$|\psi_0^{\text{rp}}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle \otimes |\psi_C\rangle \otimes |0\rangle_D. \quad (\text{B22})$$

The single-qubit factors on A, B, C are normalized complex Gaussian vectors generated with NumPy seed 42. This fixed product state is reused across the main 4-qubit crossover, early-time mechanism, and associator-null scans.

As a boundary-condition control, we also use a Haar-random pure state on the full 4-qubit Hilbert space,

$$|\psi_0^{\text{Haar}}\rangle \in \mathbb{C}^{16}, \quad (\text{B23})$$

generated as a normalized complex Gaussian vector with seed 42.

For the 4-qubit initial-state ensemble scan, we sample

$$|\psi_0^{(n)}\rangle = |\psi_A^{(n)}\rangle \otimes |\psi_B^{(n)}\rangle \otimes |\psi_C^{(n)}\rangle \otimes |0\rangle_D. \quad (\text{B24})$$

The ensemble uses NumPy seed 2027, contains 50 random product states, and is reused at every sampled time.

For the 5-qubit tetrahedral benchmark, the default initial condition is

$$|\psi_0^{(5Q)}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle \otimes |\psi_C\rangle \otimes |\psi_D\rangle \otimes |0\rangle_E. \quad (\text{B25})$$

The four internal factors are normalized complex Gaussian vectors generated with NumPy seed 123. This fixed state is used in the main tetrahedral time-resolved scan.

The explicit single-qubit amplitudes are fully determined by the NumPy seeds specified above (seed 42 for 4Q, seed 123 for 5Q) and are recorded in the accompanying code repository.

4. Scan grids

All XYZ parameter scans use uniform angular grids over $(\theta, \phi) \in [\theta_{\min}, \pi/2] \times [0, \pi/2]$ with $\theta_{\min} = 0.1$ to avoid the degenerate pole. The 4-qubit scans use a 20×20 grid (400 Hamiltonian points per time slice); the 5-qubit tetrahedral scans use a 12×12 grid (144 points).

Table I summarizes the non-angular scan axes. Time grids are piecewise uniform with denser sampling in the generative window; exact breakpoints are specified in the accompanying code.

TABLE I. Scan configurations. All 4Q scans use the 20×20 grid; the 5Q scan uses the 12×12 grid.

scan	sweep axis	range
4Q time-resolved	t (25 points)	$[0.02, 1.00]$
4Q associator-null	$\alpha \times t$	$\{0, 0.1, \dots, 1.0\} \times \{0.10, 0.15, 0.25\}$
4Q initial-state ensemble	t (11 points)	$[0.10, 1.00]$
5Q tetrahedral	t (40 points), $g_E = 0.20$	$[0.05, 1.00]$

5. Time evolution and associator-null interpolation

For all closed-system scans, the full system is evolved by exact matrix exponentiation,

$$U(t) = e^{-iHt}, \quad \rho(t) = U(t)\rho_0 U^\dagger(t). \quad (\text{B26})$$

The Hamiltonian is represented as a 16×16 matrix in the 4-qubit system and as a 32×32 matrix in the 5-qubit system. After full-system evolution, the auxiliary node is traced out before internal observables are evaluated:

$$\rho_{ABC}(t) = \text{Tr}_D[\rho(t)], \quad \rho_{ABCD}(t) = \text{Tr}_E[\rho(t)]. \quad (\text{B27})$$

At each sampled time t , the full (θ, ϕ) Hamiltonian grid is scanned; for every grid point we evolve the state, trace out the auxiliary node, and compute I_3^- , J_ρ , and J_{tot} on the reduced internal state. The Pearson correlations $r_\rho(t)$ and $r_{\text{tot}}(t)$ are then evaluated across the grid. The same procedure is used for all closed-system scans; in the tetrahedral extension,

observables are computed on the four triangular faces of $\rho_{ABCD}(t)$ and face-averaged when required.

For the open-system robustness tests, coherent evolution is first used as a preparation stage,

$$\rho_{\text{prep}} = e^{-iHt_{\text{prep}}} \rho_0 e^{iHt_{\text{prep}}}. \quad (\text{B28})$$

Starting from ρ_{prep} , the state is evolved with a Lindblad master equation,

$$\dot{\rho} = -i[H, \rho] + \sum_{\ell} \left(L_{\ell} \rho L_{\ell}^{\dagger} - \frac{1}{2} \{ L_{\ell}^{\dagger} L_{\ell}, \rho \} \right), \quad (\text{B29})$$

implemented using QuTiP's `mesolve`. In the 4-qubit robustness tests, collapse operators act on A, B, C ; in the 5-qubit tests, they act on A, B, C, D . The auxiliary nodes D and E are not directly decohered.

The three decoherence channels are implemented in Lindblad form as

$$L_q = \sqrt{\gamma} \sigma_q^-, \quad L_q = \sqrt{\gamma} \sigma_q^z, \quad L_{q,\mu} = \sqrt{\gamma/3} \sigma_q^{\mu}, \quad \mu = x, y, z, \quad (\text{B30})$$

for amplitude damping, dephasing, and depolarizing noise, respectively. The default robustness parameters are

$$t_{\text{prep}} = 0.80, \quad t_{\text{open}} = 1.20, \quad \gamma = 0.20. \quad (\text{B31})$$

Prepared families are selected using the product of I_3^- and a dyadic negativity baseline at t_{prep} . This selection is used only to retain families with both a visible tripartite signal and a nonzero dyadic entanglement background. Dyadic baselines are reported as edge means; in the triangle,

$$\bar{N}_2^{\Delta} = \frac{1}{3} N_{\text{pair, sum}}, \quad \bar{I}_2^{\Delta} = \frac{1}{3} P_{\text{pair}}, \quad (\text{B32})$$

and in the tetrahedron the corresponding means are taken over the six internal edges.

The associator-null family is constructed by removing directional incompatibility among the XYZ edge vectors while preserving their norms. For each edge ij , let

$$\mathbf{J}_{ij}^{(1)} = \mathbf{J}_{ij}(\theta, \phi), \quad \|\mathbf{J}_{ij}^{(1)}\| = R_{ij}, \quad (\text{B33})$$

and choose the common-axis reference

$$\hat{\mathbf{n}} = \hat{\mathbf{z}}, \quad \mathbf{J}_{ij}^{(0)} = R_{ij} \hat{\mathbf{n}}. \quad (\text{B34})$$

The interpolation is performed edge by edge,

$$\tilde{\mathbf{J}}_{ij}(\alpha) = (1 - \alpha)\mathbf{J}_{ij}^{(0)} + \alpha\mathbf{J}_{ij}^{(1)}, \quad \mathbf{J}_{ij}(\alpha) = R_{ij} \frac{\tilde{\mathbf{J}}_{ij}(\alpha)}{\|\tilde{\mathbf{J}}_{ij}(\alpha)\|}. \quad (\text{B35})$$

Thus α changes the directional incompatibility of the edge Hamiltonians without changing their coupling norms. The endpoint $\alpha = 0$ gives the common-axis family, while $\alpha = 1$ recovers the original fixed-norm XYZ family.

In the 4-qubit ablation scan, this interpolation is applied to AB, BC, AC , and CD . The associator-null condition concerns the internal triangle ABC : at $\alpha = 0$, the three internal edge Hamiltonians are all aligned along Z and commute pairwise, so

$$\mathcal{A}_{AB} = \mathcal{A}_{BC} = \mathcal{A}_{AC} = 0. \quad (\text{B36})$$

The auxiliary edge CD remains weak throughout the interpolation because its norm is fixed to $g_D R$; it participates in the full unitary evolution but does not enter the internal triangular associator.

Appendix C: Competition metric analysis

1. Candidate diagnostics and time-resolved comparison

To test whether the associator diagnostics provide information beyond standard state-level and pairwise quantities, we compare J_ρ and J_{tot} against a set of candidate indicators computed on the same Hamiltonian scans. All correlations in this subsection are computed across the (θ, ϕ) grid at fixed time, using the same fixed random product initial state as in the main 4-qubit benchmark.

The candidate diagnostics are summarized in Table II. They include reduced-state mixedness, pairwise correlation and entanglement baselines, a normalized associator activation ratio, a GHZ-targeted witness, and tripartite negativity proxies based on the three bipartitions.

For each candidate metric M , we compute the Pearson correlation

$$r_M(t) = \text{corr}_{(\theta, \phi)}(M(\theta, \phi; t), I_3^-(\theta, \phi; t)) \quad (\text{C1})$$

at fixed time t . The competition heatmap reports these correlations for

$$t \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50, 0.75, 1.00\}. \quad (\text{C2})$$

TABLE II. Candidate diagnostics compared with I_3^- in the time-resolved competition analysis.

diagnostic	definition	role
J_ρ	state-resolved associator readout	associator activation
J_{tot}	bare associator capacity	Hamiltonian-level capacity
η	J_ρ/J_{tot}	normalized activation efficiency
S_L	$1 - \text{Tr}[\rho_{ABC}^2]$	reduced-state mixedness
P_{pair}	$I(A : B) + I(A : C) + I(B : C)$	pairwise total correlation
N_{sum}	$\mathcal{N}_{AB} + \mathcal{N}_{AC} + \mathcal{N}_{BC}$	pairwise entanglement
W_{GHZ}	$\langle \text{GHZ} \rho_{ABC} \text{GHZ} \rangle - \frac{1}{2}$	GHZ-targeted witness
$N_{\text{tri,min}}$	$\min\{\mathcal{N}_{A BC}, \mathcal{N}_{B AC}, \mathcal{N}_{C AB}\}$	bipartition inseparability
$N_{\text{tri,geom}}$	$(\mathcal{N}_{A BC} \mathcal{N}_{B AC} \mathcal{N}_{C AB})^{1/3}$	bipartition inseparability

The main heatmap includes J_ρ , J_{tot} , η , P_{pair} , N_{sum} , W_{GHZ} , and S_L . The tripartite negativity proxies are analyzed separately as diagnostic checks, because their behavior is dominated by bipartition inseparability rather than by the negative tripartite-information sector.

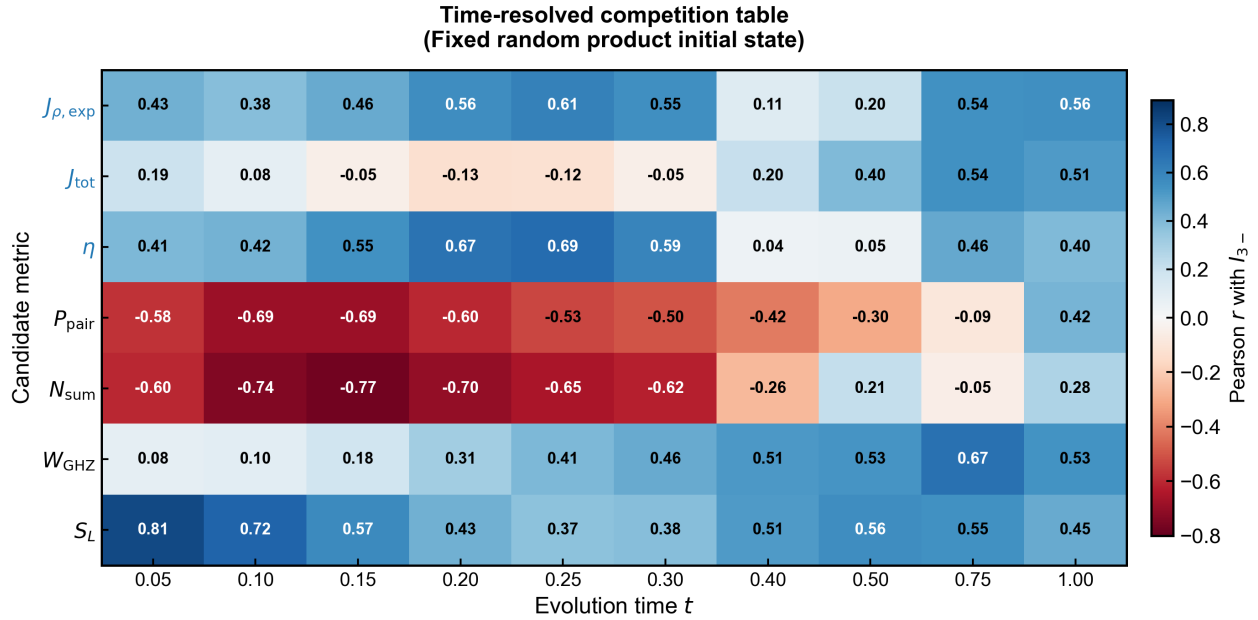


FIG. 5. Time-resolved diagnostic competition in the 4-qubit triangle network. For each metric M , the heatmap reports the Pearson correlation between M and I_3^- across the (θ, ϕ) Hamiltonian grid at fixed time.

The heatmap confirms the two-layer structure reported in the main text: J_ρ leads in the early activation window, while J_{tot} becomes the strongest structural predictor at later times. Among the alternatives, S_L captures a necessary mixedness threshold but not the specific Hamiltonian-level channel; pairwise baselines and the GHZ witness track lower-order or class-specific features that do not selectively isolate the synergy-dominant sector. The tripartite negativity proxies characterize bipartition inseparability rather than the information balance measured by I_3^- . The associator diagnostics are the only candidates tied directly to the three-edge incompatibility identified in Sec. A.

2. Associator-null recovery hierarchy

The associator-null interpolation provides a second metric comparison: instead of asking which diagnostic correlates with I_3^- at a fixed Hamiltonian family, it asks how different quantities recover as the non-associative channel is restored from the exact null manifold.

For each scan point (θ, ϕ) , time t , and interpolation strength α , we compute the normalized recovery ratios

$$R_{I_3^-}(\alpha; t) = \frac{I_3^-(\alpha; t)}{I_3^-(1; t)}, \quad (\text{C3})$$

and

$$R_{J_\rho}(\alpha; t) = \frac{J_\rho(\alpha; t)}{J_\rho(1; t)}, \quad R_{J_{\text{tot}}}(\alpha) = \frac{J_{\text{tot}}(\alpha)}{J_{\text{tot}}(1)}. \quad (\text{C4})$$

Ratios are evaluated pointwise on the Hamiltonian grid and then summarized by the median and interquartile range. Grid points whose $\alpha = 1$ denominator is below numerical tolerance are excluded from the corresponding ratio summary.

The recovery curves separate the two associator layers. The bare capacity J_{tot} rises rapidly as soon as $\alpha > 0$, because the edge Hamiltonians quickly regain noncommuting XYZ components. However, the tripartite synergy indicator I_3^- does not recover at the same rate. Its recovery is slower and is tracked more closely by J_ρ , indicating that restoring operator-level non-associative capacity is not sufficient to immediately restore the negative tripartite-information signal.

This recovery hierarchy is the interpolation-side counterpart of the time-resolved capacity activation distinction. J_{tot} measures the restored Hamiltonian-level associator capacity, while J_ρ measures state-conditioned access to that capacity. The slower recovery of I_3^-

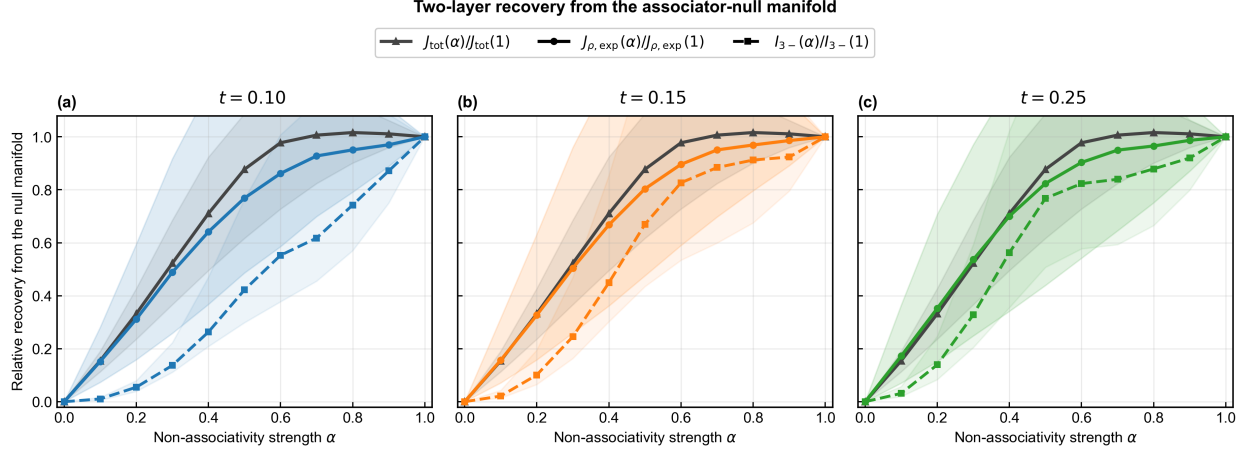


FIG. 6. Associator-null recovery hierarchy. The bare capacity J_{tot} recovers rapidly once the edge Hamiltonians leave the common-axis null manifold, whereas I_3^- recovers more slowly and follows the state-resolved diagnostic J_ρ more closely.

therefore supports the interpretation that state activation is the bottleneck between available non-associative structure and the observable tripartite-information signal.

The pairwise baselines provide an additional control. Along the same interpolation, pairwise mutual information and pairwise negativity remain finite at the null point and reorganize as α is increased, but they do not exhibit the same null collapse as I_3^- . Thus the recovery analysis distinguishes three roles: J_{tot} tracks restored algebraic capacity, J_ρ tracks state-resolved activation, and I_3^- records when the reduced tripartite state enters the synergy-dominant information sector.

Appendix D: Initial-state robustness and boundary conditions

1. Boundary conditions: plus and Haar-random starts

The main benchmark uses a fixed random product initial state. To delimit the validity domain of the associator diagnostics, we also test two boundary conditions: an axis-biased product state,

$$|\psi_0^+\rangle = |+\rangle_A \otimes |+\rangle_B \otimes |+\rangle_C \otimes |0\rangle_D, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad (\text{D1})$$

and a Haar-random pure state on the full 4-qubit Hilbert space,

$$|\psi_0^{\text{Haar}}\rangle \in \mathbb{C}^{16}. \quad (\text{D2})$$

The $|+\rangle^{\otimes 3} \otimes |0\rangle$ control preserves a fully unentangled start and produces a strong early association between J_ρ and I_3^- . However, the response is more transient and more strongly axis-dependent than in the fixed random product benchmark. We therefore use this state as an onset control rather than as the main reference case.

The Haar-random control probes the opposite boundary. Because the system starts from a highly entangled state with already mixed reduced subsystems, the scan no longer isolates the clean generation of tripartite structure from a factorized state. In this regime the associator diagnostics lose their role as sharp onset coordinates, and the parameter-space structure is dominated by the pre-existing many-body content of the initial state.

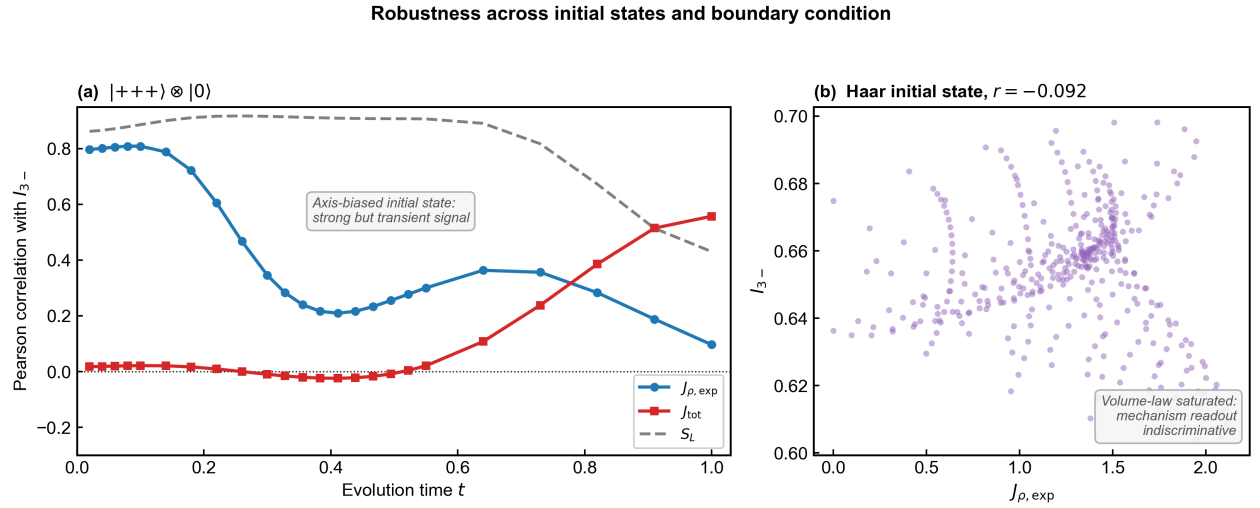


FIG. 7. Boundary-condition tests for the 4-qubit triangle network. The plus-state control gives a strong but axis-biased transient response, while the Haar-random control starts from a highly entangled state and therefore no longer probes clean dynamical generation from a product state.

These controls delimit the intended scope of the method. The associator diagnostics are most informative in low-entanglement generative settings, where the negative tripartite-information signal is created dynamically by the interaction Hamiltonian. They are not intended as a universal classifier for arbitrary quantum states.

For reference, the Page-formula estimate for a Haar-random 4-qubit pure state traced to three qubits gives

$$\mathbb{E}[I_3] = 4 \mathbb{E}[S_A] - 3 \mathbb{E}[S_{AB}] \approx -0.528, \quad (\text{D3})$$

corresponding to $\mathbb{E}[I_3^-] \approx 0.528$. A numerical ensemble of 10000 Haar samples gives

$$\mathbb{E}[I_3^-] = 0.528 \pm 0.175, \quad (\text{D4})$$

with 99.7% of samples in the synergy-dominant regime. This value is used only as a Haar baseline; it is not part of the clean generative benchmark.

2. Random-product ensemble scan

We next test how the state-resolved diagnostic behaves across an ensemble of admissible low-entanglement starts. We sample 50 random product initial states,

$$|\psi_0^{(n)}\rangle = |\psi_A^{(n)}\rangle \otimes |\psi_B^{(n)}\rangle \otimes |\psi_C^{(n)}\rangle \otimes |0\rangle_D, \quad n = 1, \dots, 50. \quad (\text{D5})$$

For each initial state and each sampled time, the same 20×20 fixed-norm XYZ Hamiltonian grid is rescanned. This separates variation across Hamiltonians from variation across initial product states.

For each state n and time t , we compute the within-state diagnostic correlations and the strongest tripartite synergy indicator generated by that state:

$$r_\rho^{(n)}(t) = \text{corr}_{(\theta, \phi)} \left(J_\rho^{(n)}(\theta, \phi; t), I_3^{-, (n)}(\theta, \phi; t) \right), \quad (\text{D6})$$

$$r_{\text{tot}}^{(n)}(t) = \text{corr}_{(\theta, \phi)} \left(J_{\text{tot}}(\theta, \phi), I_3^{-, (n)}(\theta, \phi; t) \right), \quad (\text{D7})$$

$$I_{3, \text{max}}^{-, (n)}(t) = \max_{(\theta, \phi)} I_3^{-, (n)}(\theta, \phi; t). \quad (\text{D8})$$

The ensemble summaries report medians and interquartile ranges of these quantities across the 50 initial states.

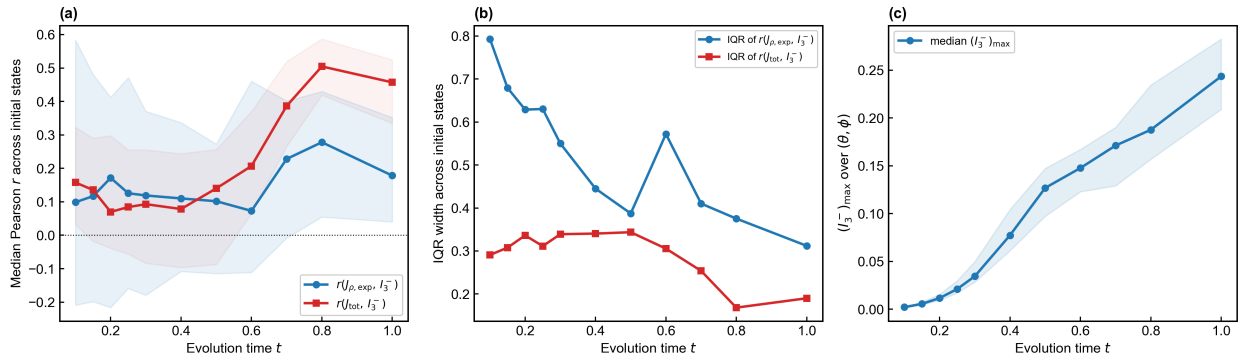


FIG. 8. Random-product ensemble scan in the 4-qubit triangle. The panels show the ensemble-level behavior of the within-state diagnostic correlations and the maximum generated I_3^- across 50 random product starts.

The ensemble scan shows that state activation has an effective domain: some product states enter the early activation window much more strongly than others. At early times, states that generate larger $I_{3,\max}^-$ also tend to show a higher within-state quality of the J_ρ diagnostic. Thus J_ρ should be interpreted as a state-conditioned readout of the associator sector, not as a universal scalar predictor that works equally well for every initial state.

The same ensemble confirms the late-time stability of the bare capacity diagnostic. While J_ρ is more sensitive to initial-state alignment at early times, J_{tot} becomes more stable across the random-product ensemble at later times. This supports the two-layer picture: early behavior is access-sensitive, whereas later behavior increasingly reveals the state-independent associator capacity encoded in the Hamiltonian.

Appendix E: Transverse-field Ising model validation

1. Analytic associator and pure-Ising null

We first consider a transverse-field Ising (TF-Ising) realization of the triangle network. For each internal edge, we write

$$H_{ij}^{\text{TF}} = J_{ij}Z_iZ_j + \frac{h}{2}(X_i + X_j), \quad (\text{E1})$$

so that the sum over the three triangle edges gives

$$H_{\Delta}^{\text{TF}} = J_{AB}Z_AZ_B + J_{BC}Z_BZ_C + J_{AC}Z_AZ_C + h(X_A + X_B + X_C). \quad (\text{E2})$$

The factor $1/2$ in the edge-resolved expression avoids double-counting the transverse field, since each triangle node appears in two internal edges.

Using the mediator-resolved associators defined in Sec. A, direct Pauli algebra gives

$$J_{\text{tot}}^2 = h^4 \sum_{(ij) \in \{AB, BC, AC\}} J_{ij}^2. \quad (\text{E3})$$

Under the fixed-norm parametrization $J_{AB}^2 + J_{BC}^2 + J_{AC}^2 = R^2$, this reduces to

$$J_{\text{tot}} = h^2 R. \quad (\text{E4})$$

Thus the TF-Ising associator is activated by the transverse field and vanishes in the pure Ising limit $h \rightarrow 0$.

The pure Ising limit provides a textbook associator-null control. At $h = 0$,

$$H_{\Delta}^{\text{Ising}} = J_{AB}Z_AZ_B + J_{BC}Z_BZ_C + J_{AC}Z_AZ_C, \quad (\text{E5})$$

so all internal edge Hamiltonians commute pairwise. Hence $J_{\text{tot}} = 0$, and the Cauchy–Schwarz bound in Sec. A 3 gives $J_{\rho} = 0$ for every reduced triangle state.

Numerically, the pure Ising scan confirms this null prediction. Across the full edge-geometry grid and sampled evolution times, the tripartite synergy indicator remains at the numerical floor $I_3^- \simeq 0$, while lower-order pairwise observables can remain nonzero. Thus the absence of I_3^- is not caused by removing pairwise interactions; it follows from eliminating the non-associative channel. This verifies in the TF-Ising setting the same null structure tested by the associator-null interpolation in the XYZ family.

2. Edge-geometry dependence

We next test how the distribution of fixed Ising strength over the three triangle edges affects the conversion of an activated associator channel into the tripartite synergy indicator. The edge magnitudes are parametrized as

$$K_{AB} = R \sin \theta \cos \phi, \quad K_{BC} = R \sin \theta \sin \phi, \quad K_{AC} = R \cos \theta, \quad (\text{E6})$$

so that

$$K_{AB}^2 + K_{BC}^2 + K_{AC}^2 = R^2. \quad (\text{E7})$$

At fixed transverse field h , every point in the (θ, ϕ) scan therefore has the same bare associator capacity $J_{\text{tot}} = h^2 R$. The scan is thus a direct test of edge-geometry effects at fixed non-associative capacity. The representative trajectories and geometry landscape shown below use the near-competition ferromagnetic regime

$$\lambda = 1.00, \quad J_{AB}, J_{BC}, J_{AC} < 0. \quad (\text{E8})$$

Figure 9 compares representative trajectories for three edge-strength geometries. In near-complete triangles, J_{ρ} opens early and I_3^- grows later, consistent with the activation picture. By contrast, isolated-dimer and open-wedge geometries can still show appreciable J_{ρ} while producing only weak I_3^- . Thus a large state-activated associator readout is not by itself sufficient to generate a large entropy-level tripartite signal.

To summarize this effect across the full geometry scan, we compute the time-integrated quantities

$$A_{I_3}(\theta, \phi) = \int I_3^-(t; \theta, \phi) dt, \quad A_{J_\rho}(\theta, \phi) = \int J_\rho(t; \theta, \phi) dt. \quad (\text{E9})$$

As shown in Fig. 10, high A_{J_ρ} does not automatically imply high A_{I_3} . The geometry dependence is partly organized by the normalized edge weights

$$w_{ij} = \frac{K_{ij}^2}{K_{AB}^2 + K_{BC}^2 + K_{AC}^2}, \quad (\text{E10})$$

and by the triangle-closure factor

$$C_\Delta = 27 w_{AB} w_{BC} w_{AC}. \quad (\text{E11})$$

This factor vanishes when any edge is absent and equals one for a perfectly balanced triangle. Although it is not a complete predictor of I_3^- , it captures the geometric penalty associated with nearly broken triangular support.

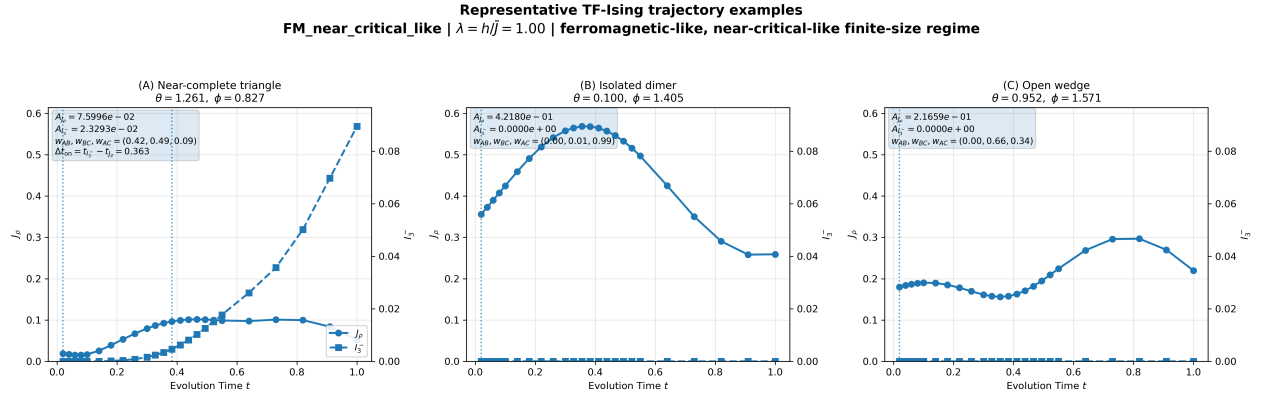


FIG. 9. Representative TF-Ising trajectories for three edge-strength geometries at fixed bare associator capacity.

These results show that non-associative capacity $J_{\text{tot}} > 0$ and state activation $J_\rho > 0$ are individually necessary but jointly insufficient: the tripartite synergy indicator additionally requires effective triangular closure, i.e., all three edges must carry substantial coupling weight. This geometric constraint is generically satisfied in the XYZ family used in the main text, but becomes explicit in the TF-Ising family, where the spherical parametrization allows individual edges to approach zero.

Geometry dependence of tripartite synergy indicator in the TF-Ising triangle
FM_near_critical_like | $\lambda = h/\bar{J} = 1.00$ | ferromagnetic-like, near-critical-like finite-size regime

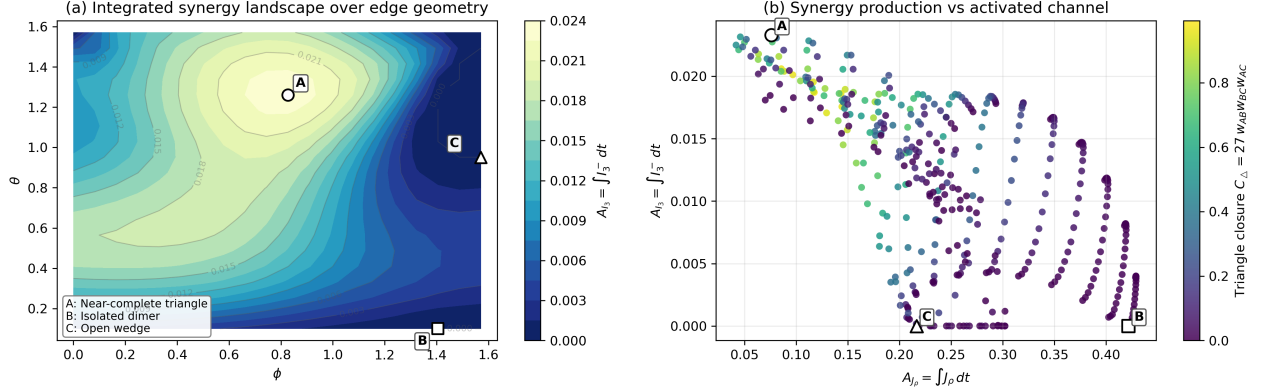


FIG. 10. Geometry dependence of the integrated tripartite synergy signal in the TF-Ising triangle.

3. Field-interaction scan

We next scan the transverse-field strength relative to the RMS Ising coupling,

$$\lambda = \frac{h}{\bar{J}}. \quad (\text{E12})$$

For each λ , the same fixed-norm edge-geometry grid is scanned over time for both uniform sign choices,

$$J_{AB}, J_{BC}, J_{AC} < 0 \quad (\text{FM-like}), \quad J_{AB}, J_{BC}, J_{AC} > 0 \quad (\text{AFM-like}). \quad (\text{E13})$$

The refined scan uses $\lambda \in [0, 3]$, $\Delta\lambda = 0.05$, with 28 time points in $t \in [0.02, 3.00]$.

We summarize the finite-time response using

$$M_{\max}(\lambda) = \max_{t, \theta, \phi} I_3^-(t; \theta, \phi), \quad (\text{E14})$$

$$M_{\text{mean}}(\lambda) = \max_t \langle I_3^-(t; \theta, \phi) \rangle_{\theta, \phi}, \quad (\text{E15})$$

$$A_{\text{mean}}(\lambda) = \int \langle I_3^-(t; \theta, \phi) \rangle_{\theta, \phi} dt. \quad (\text{E16})$$

Here $\langle \cdot \rangle_{\theta, \phi}$ denotes the average over the edge-geometry grid.

Figure 11 shows a nonmonotonic response. In the interaction-dominated limit $\lambda \ll 1$, the transverse field is too weak to activate the non-associative channel efficiently. In the field-dominated limit $\lambda \gg 1$, local rotations dominate and the activated channel is not efficiently converted into a tripartite synergy signal. The strongest response therefore occurs in an intermediate field–interaction competition window.

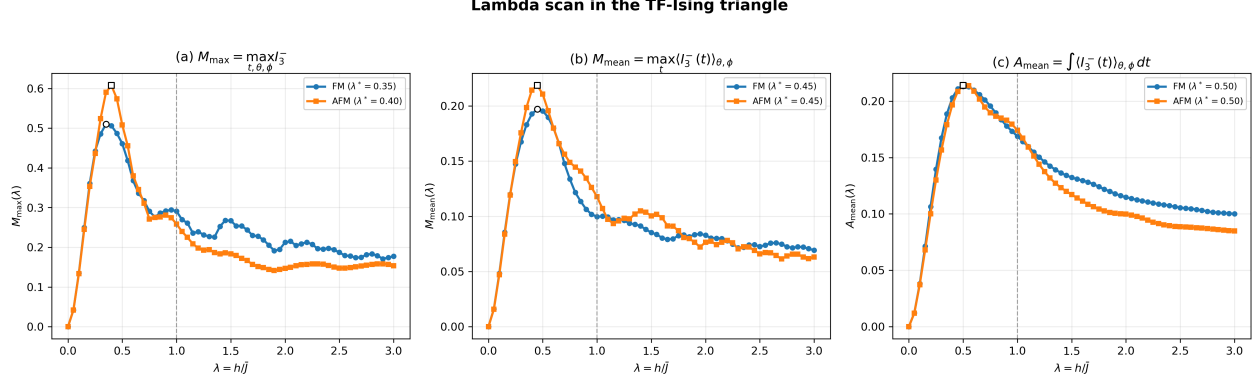


FIG. 11. Field-interaction scan in the TF-Ising triangle. The panels show the optimal peak $M_{\max}$, the geometry-averaged peak M_{mean} , and the time-integrated mean production A_{mean} as functions of $\lambda = h/\bar{J}$ for FM-like and AFM-like sign choices.

The precise optimum depends on the finite-window summary. The peak statistics $M_{\max}$ and M_{mean} are maximized on the interaction-dominated side of the competition window, while the integrated response A_{mean} is enhanced over a broader intermediate range. We therefore interpret the scan as evidence for a finite-size competition window, not as a sharp critical-point maximum. The robust conclusion is that I_3^- generation is suppressed in both extreme limits and is enhanced when transverse-field and Ising scales compete.

4. Frustration and the Edwards–Anderson sign ensemble

We finally test whether sign frustration enhances or suppresses tripartite-synergy generation at fixed coupling magnitudes. For each edge-geometry point (θ, ϕ) and field strength λ , we enumerate all eight sign patterns

$$(s_{AB}, s_{BC}, s_{AC}) \in \{-1, +1\}^3, \quad J_{ij} = s_{ij} K_{ij}. \quad (\text{E17})$$

In our convention, $J_{ij} < 0$ is ferromagnetic-like and $J_{ij} > 0$ is antiferromagnetic-like. The frustration classification is

$$\text{sign}(J_{AB}J_{BC}J_{AC}) \begin{cases} > 0, & \text{frustrated,} \\ < 0, & \text{non-frustrated.} \end{cases} \quad (\text{E18})$$

The scan uses an interior 16×16 geometry grid, six field strengths $\lambda \in \{0.50, 0.75, 1.00, 1.25, 1.50, 2.00\}$, and ten evolution times in $t \in [0.05, 2.00]$. For each fixed (λ, θ, ϕ) , all eight sign patterns

share the same edge magnitudes and therefore the same bare capacity,

$$J_{\text{tot}}^2 = h^4 \sum_{(ij)} J_{ij}^2. \quad (\text{E19})$$

The scan therefore isolates the effect of sign structure and frustration from the effect of Hamiltonian-level associator capacity.

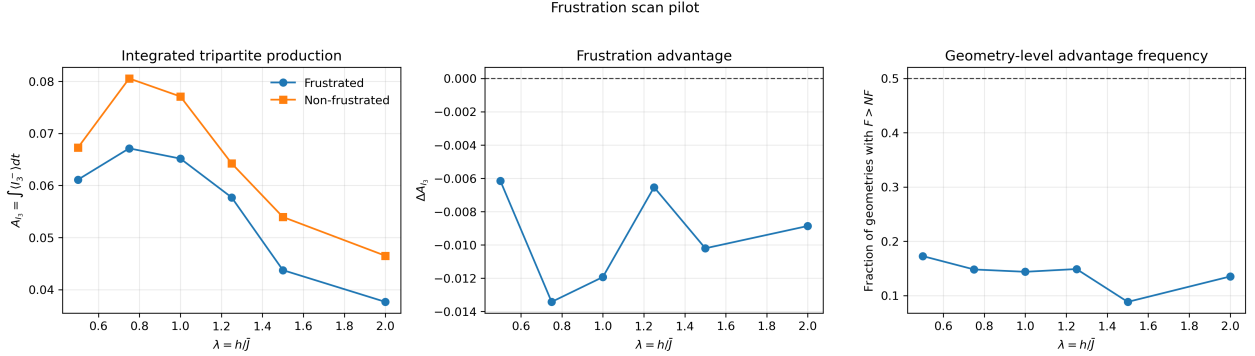


FIG. 12. Edwards–Anderson sign ensemble at fixed edge magnitudes. For each fixed (λ, θ, ϕ) , all sign patterns have identical J_{tot} . Frustrated configurations produce systematically lower integrated I_3^- than non-frustrated configurations, with only a minority of geometries showing the reverse trend.

Across the scanned λ values and edge geometries, frustrated sign configurations generate less integrated I_3^- than their non-frustrated counterparts, with only 10 ~ 18% of geometries showing the reverse trend. Since J_{tot} is identical across sign patterns at fixed edge magnitudes, this suppression cannot be attributed to the bare associator capacity. It instead reflects the state-activated layer and subsequent coherent dynamics: frustration reduces the efficiency with which the evolving state accesses the available non-associative sector.

This result should be distinguished from equilibrium studies in which frustration can enhance synergy in classical Ising models [15]. Here the question is dynamical generation from initially unentangled states, and frustration suppresses the conversion of fixed algebraic capacity into the observed tripartite-information signal.

5. Graph additivity of the TF-Ising bare capacity

The analytic TF-Ising associator gives a simple graph-level decomposition of the bare capacity. Let $G = (V, E)$ be a simple graph with qubits on vertices, and let $\mathcal{T}(G)$ denote

the set of triangular faces. For a face $f = \{a, b, c\} \in \mathcal{T}(G)$, we define the face-local probe Hamiltonians on the three-qubit Hilbert space of that face, for example

$$\tilde{H}_{ab}^{(f)} = J_{ab}Z_aZ_b + \frac{h}{2}(X_a + X_b), \quad (\text{E20})$$

with analogous definitions for $\tilde{H}_{bc}^{(f)}$ and $\tilde{H}_{ac}^{(f)}$. The corresponding mediator associators are constructed from these three face-local edge Hamiltonians as in Sec. A. Under this convention, the associator assigned to a face depends only on the three edge terms inside that face, not on how the face is embedded in the rest of the graph.

Using the single-triangle TF-Ising result, every face satisfies

$$J_{\text{tot}}^2(f) = h^4 \sum_{(ij) \in f} J_{ij}^2. \quad (\text{E21})$$

Thus the per-face TF-Ising bare capacity is independent of all other edges and vertices of G . Defining the graph-summed capacity by

$$\Sigma J_{\text{tot}}^2(G) = \sum_{f \in \mathcal{T}(G)} J_{\text{tot}}^2(f), \quad (\text{E22})$$

and the triangle multiplicity of edge (ij) by

$$m_{ij}(G) = |\{f \in \mathcal{T}(G) : (ij) \subset f\}|, \quad (\text{E23})$$

we obtain the graph-summed identity

$$\Sigma J_{\text{tot}}^2(G) = h^4 \sum_{(ij) \in E} m_{ij}(G) J_{ij}^2. \quad (\text{E24})$$

The graph dependence therefore enters only through the triangle multiplicity $m_{ij}(G)$.

For the tetrahedral core K_4 used in the 5-qubit extension, each of the six internal edges belongs to two triangular faces. With equal edge magnitude $J_{ij} = R$,

$$\Sigma J_{\text{tot}}^2(K_4) = 12h^4 R^2, \quad \langle J_{\text{tot}}^2 \rangle_{\text{face}} = 3h^4 R^2. \quad (\text{E25})$$

For $R = \sqrt{3}$ and $h = 1$, this gives $\langle J_{\text{tot}}^2 \rangle_{\text{face}} = 9$, in agreement with the numerical verification.

The same identity gives an analytic scaling law for triangle-decorated graph families. For a triangular ladder G_N with N triangular faces and uniform coupling $J_{ij} = J$,

$$\sum_{(ij) \in E} m_{ij}(G_N) = 3N, \quad (\text{E26})$$

and hence

$$\Sigma J_{\text{tot}}^2(G_N) = 3h^4 J^2 N, \quad \sqrt{\Sigma J_{\text{tot}}^2(G_N)} = \sqrt{3} h^2 J \sqrt{N}, \quad (\text{E27})$$

while the face mean remains intensive,

$$\langle J_{\text{tot}}^2 \rangle_{\text{face}} = \frac{\Sigma J_{\text{tot}}^2(G_N)}{|\mathcal{T}(G_N)|} = 3h^4 J^2. \quad (\text{E28})$$

More generally, for bounded-degree triangle-decorated graph families with $|\mathcal{T}(G_N)| \propto N$ and bounded $m_{ij}(G_N)$, the graph-summed bare capacity is extensive in the number of faces, whereas the face-mean capacity is N -independent. The relative fluctuation of a face-summed capacity is therefore expected to scale as

$$\frac{\sigma}{\mu} \sim \frac{1}{\sqrt{N}}. \quad (\text{E29})$$

This graph-additive structure provides the analytic basis for the face-averaging interpretation used in the main text. In the TF-Ising specialization, J_{tot}^2 is a face-local and additive structural capacity. What remains dynamical is whether J_ρ , and ultimately I_3^- , concentrate around the same structural prediction as the number of faces increases. The additivity result therefore supports the bare-capacity layer of the two-layer framework, while larger- N scans are needed to test whether the predictive relation between face-mean J_{tot} and face-mean I_3^- also stabilizes.

The identities above were checked numerically by explicit Pauli-matrix construction of the face-local Hamiltonians. The per-face formula, triangular-ladder identity for $N = 1, \dots, 8$, and K_4 tetrahedral case were all reproduced to machine precision. These checks verify the implementation; the identities themselves follow from the face-local definition and the single-triangle TF-Ising associator formula.

Appendix F: Open-system robustness of higher-order information

The Lindblad channels, collapse operators, preparation time, noise duration, and noise rate are specified in Sec. B 5. Here we compare the tripartite synergy indicator with dyadic mutual-information and negativity baselines after coherent preparation followed by noisy evolution.

In the 4-qubit triangle, a pool of 80 candidate Hamiltonian points is prepared unitarily, and the top $k = 12$ families are retained using the joint criterion of a visible I_3^- signal

and a nonzero dyadic entanglement baseline at the preparation time. Noise acts on the internal triangle nodes A, B, C , while the auxiliary node D is not directly decohered. The higher-order observable is I_3^- evaluated on

$$\rho_{ABC}(t) = \text{Tr}_D[\rho(t)]. \quad (\text{F1})$$

The dyadic baselines are the mean pairwise mutual information and mean pairwise negativity,

$$\bar{I}_2^\Delta = \frac{1}{3}[I(A : B) + I(A : C) + I(B : C)], \quad \bar{N}_2^\Delta = \frac{1}{3}[\mathcal{N}(A : B) + \mathcal{N}(A : C) + \mathcal{N}(B : C)]. \quad (\text{F2})$$

In the 5-qubit tetrahedral network, a pool of 60 candidate Hamiltonian points is prepared unitarily, and the top $k = 10$ families are retained using the same selection criterion applied to the face-averaged tripartite signal and the internal dyadic entanglement baseline. Noise acts on the tetrahedral nodes A, B, C, D , while the external node E is not directly decohered. The higher-order observable is the face-averaged tripartite synergy indicator,

$$\overline{I_3^-}^{\text{face}} = \frac{1}{4} \sum_{f \in \{ABC, ABD, ACD, BCD\}} I_3^-(f). \quad (\text{F3})$$

The dyadic baselines are internal-edge means over the six tetrahedral edges,

$$\bar{I}_2^{\text{int}} = \frac{1}{6} \sum_{(ij) \in E_{\text{tet}}} I(i : j), \quad \bar{N}_2^{\text{int}} = \frac{1}{6} \sum_{(ij) \in E_{\text{tet}}} \mathcal{N}(i : j), \quad (\text{F4})$$

where

$$E_{\text{tet}} = \{AB, AC, AD, BC, BD, CD\}. \quad (\text{F5})$$

Using internal-edge means avoids double-counting edges that appear in multiple triangular faces.

Figure 13 shows the same qualitative pattern in both geometries. The pairwise negativity baseline is rapidly suppressed after noise onset, as expected for an entanglement-sensitive dyadic observable. The pairwise mutual-information baseline is more persistent because it includes both classical and quantum correlations, but it does not track the higher-order indicator as a fixed proportion. Instead, the relative ratios

$$\frac{I_3^-}{\bar{I}_2^\Delta}, \quad \frac{\overline{I_3^-}^{\text{face}}}{\bar{I}_2^{\text{int}}} \quad (\text{F6})$$

increase during noisy evolution for the tested channels.

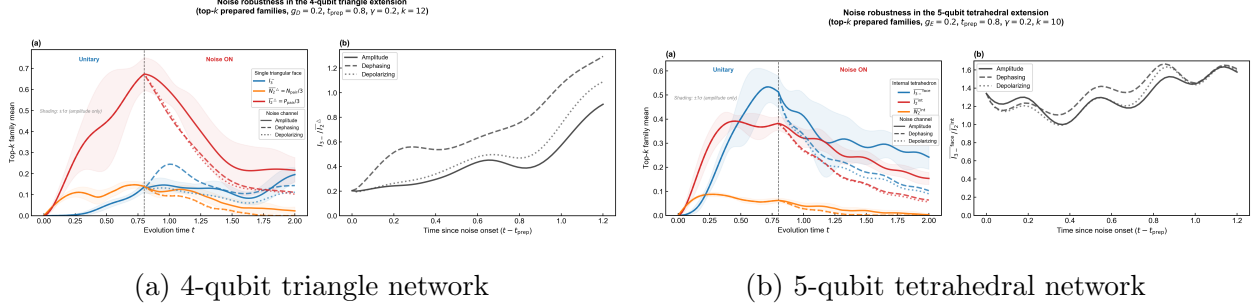


FIG. 13. Open-system robustness in the triangle and tetrahedral networks. (a) In the 4-qubit triangle, the tripartite synergy indicator I_3^- is compared with mean dyadic mutual-information and negativity baselines under amplitude damping, dephasing, and depolarizing noise. (b) In the 5-qubit tetrahedral network, the face-averaged tripartite synergy indicator $\overline{I_3^{\text{face}}}$ is compared with internal-edge dyadic baselines. In both geometries, the higher-order indicator remains visible after noise onset and retains an increasing relative share compared with the dyadic mutual-information baseline.

We therefore interpret the open-system result as relative structural persistence, not as absolute protection from decoherence. The higher-order signal can decay in magnitude, but it retains a larger share of the remaining reduced-state correlation structure than the dyadic baselines. The same trend appears under two different normalizations: one tripartite face versus three edges in the triangle, and four tripartite faces versus six internal edges in the tetrahedron. Thus the robustness effect is not an artifact of the minimal triangle geometry.

Appendix G: Supplementary results for the tetrahedral network

This section collects supplementary results specific to the 5-qubit tetrahedral extension. The Hamiltonian construction, initial state, scan grid, and external-coupling convention are specified in Sec. B. The time-resolved face-averaged diagnostics are presented in Fig. 4 of the main text.

1. Environment-edge control in the tetrahedral network

To check whether the tetrahedral higher-order response can be reduced to correlations on the external coupling edge, we monitor observables on the edge DE . These quantities

are computed from the full 5-qubit state before tracing out the external node:

$$\rho_{DE}(t) = \text{Tr}_{ABC}[\rho(t)]. \quad (\text{G1})$$

We evaluate the environment-edge mutual information and negativity,

$$I_{DE}(t) = S_D(t) + S_E(t) - S_{DE}(t), \quad N_{DE}(t) = \frac{\|\rho_{DE}^{T_D}(t)\|_1 - 1}{2}. \quad (\text{G2})$$

These observables are not part of the internal tetrahedral average. They are used only to diagnose how strongly the external node becomes correlated or entangled with the tetrahedron through the edge DE .

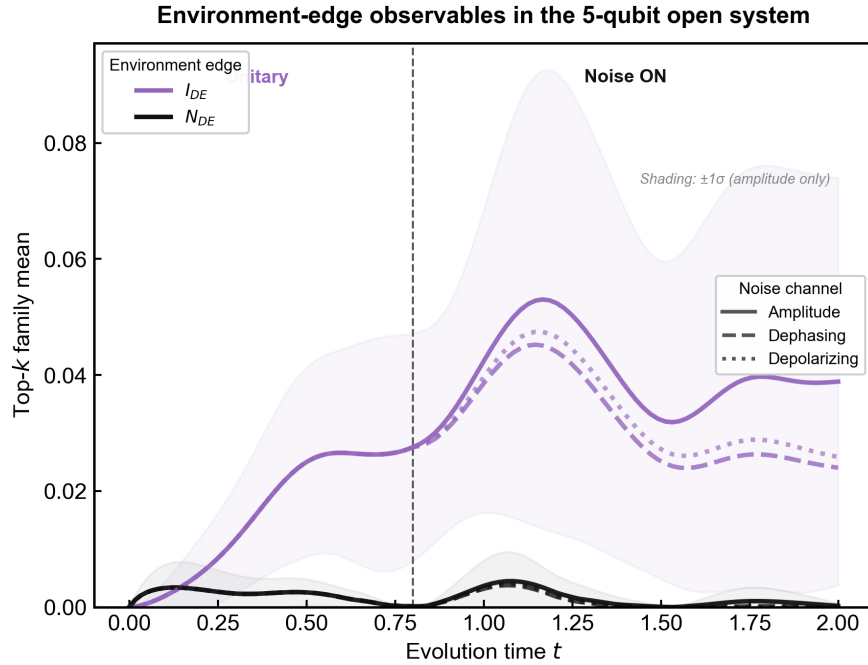


FIG. 14. Environment-edge observables in the 5-qubit open-system test. The mutual information I_{DE} and negativity N_{DE} monitor correlations and entanglement on the external coupling edge. Their trajectories do not simply reproduce the internal face-averaged tripartite synergy response.

During the unitary preparation stage, I_{DE} and N_{DE} quantify how much correlation and entanglement are carried by the external edge before decoherence is applied. In the open-system stage, noise acts on the internal tetrahedral nodes A, B, C, D , but not directly on E . The edge DE therefore provides a monitor of how external-edge correlations respond indirectly to internal decoherence.

Figure 14 shows that the environment-edge observables do not simply mirror the internal face-averaged tripartite synergy indicator. In particular, the internal higher-order signal can

remain visible even when N_{DE} is small or decaying. Thus the tetrahedral robustness result cannot be reduced to a monotonic buildup of edge-local entanglement on DE .

The external edge acts as a coupling channel that reshapes the reduced tetrahedral dynamics, but the observed higher-order response remains an internal face-level property. This supports the same conclusion as the g_E -scan: external coupling redistributes state activation across tetrahedral faces, but it does not replace the internal associator-based mechanism.

2. Environment-coupling scan and topological differentiation

For each triangular face $f \in \{ABC, ABD, ACD, BCD\}$ of the reduced state $\rho_{ABCD}(t) = \text{Tr}_E[\rho(t)]$, we compute $I_3^-(f)$, $J_\rho(f)$, and $J_{\text{tot}}(f)$; tetrahedral summaries use arithmetic face means. We next examine how the tetrahedral higher-order response changes as the coupling to the external node is varied. We fix a representative high-signal Hamiltonian point,

$$\theta^* = 0.501126, \quad \phi^* = 0.856798, \quad t^* = 0.20, \quad (\text{G3})$$

and vary the external-edge strength as

$$\|\mathbf{J}_{DE}\| = g_E R, \quad g_E \in \{0.05, 0.10, 0.20, 0.30, 0.50, 0.80, 1.00, 1.50, 2.00\}. \quad (\text{G4})$$

For each g_E , the full 5-qubit state is evolved unitarily, the external node E is traced out, and the four triangular faces of the reduced tetrahedron are evaluated separately.

Because E couples through node D , the face ABC is distal to the environment edge, whereas ABD , ACD , and BCD are adjacent to it. We therefore define

$$I_{3,\text{distal}}^- = I_3^-(ABC), \quad I_{3,\text{adj}}^- = \frac{1}{3} [I_3^-(ABD) + I_3^-(ACD) + I_3^-(BCD)], \quad (\text{G5})$$

and the topological gap

$$\Delta_{\text{topo}} = I_{3,\text{distal}}^- - I_{3,\text{adj}}^-. \quad (\text{G6})$$

A positive Δ_{topo} means that the face not containing the environment-coupled node D carries a stronger tripartite synergy indicator than the environment-adjacent faces.

Figure 15 shows that increasing g_E does not produce a monotonic enhancement of the tetrahedral signal. The face-averaged tripartite synergy indicator first increases slightly, peaks near $g_E = 0.50$, and then decreases at stronger coupling. The distal face ABC peaks later, near $g_E = 0.80$, and remains larger than the adjacent-face mean as g_E increases.

Figure 5: Environment-coupling scan on the best 5-qubit tetrahedral point
($\theta = 0.501$, $\phi = 0.857$, $t = 0.20$)

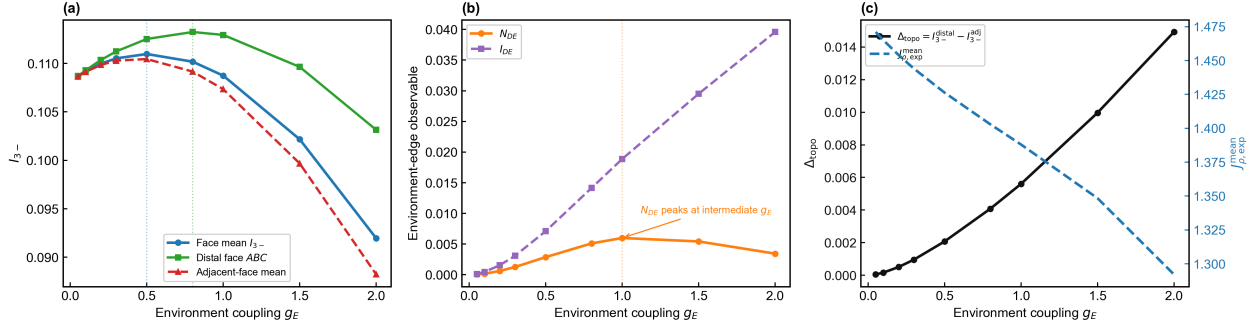


FIG. 15. Environment-coupling scan in the 5-qubit tetrahedral network. Varying g_E does not uniformly enhance the tetrahedral response; instead it differentiates the distal face ABC from the three environment-adjacent faces.

Consequently, Δ_{topo} grows from nearly zero at weak coupling to a clearly positive value at strong coupling.

This face-level differentiation occurs without changing the internal bare capacity. Along the g_E scan, the mean J_{tot} of the internal faces remains fixed because the internal tetrahedral Hamiltonian is unchanged. By contrast, the mean state-resolved diagnostic J_{ρ} decreases as g_E grows. Thus the external edge does not modify the available internal non-associative capacity; it changes how the evolving state accesses the face-resolved associator sectors.

The associated environment-edge observables show a different trend: I_{DE} generally increases with g_E , whereas N_{DE} peaks at intermediate coupling and then decreases. Thus stronger edge-local correlation with the external node does not imply a monotonic increase of the internal tripartite synergy indicator. The g_E -scan therefore supports a topology-sensitive version of the capacity–activation picture: external coupling redistributes state activation across tetrahedral faces while leaving the internal associator capacity unchanged.

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